engineering 4 us all

Student Handout

Name: ______________________________ **Date:** _________________ **Class:** _______________

MyDesign Portfolio

Element B Initial Template

Element B: Documentation and Analysis of Prior Design Attempts

https://www.sciencedirect.com/science/article/pii/S0165993618305247 ←helpful source

Prior Design Attempt #1 - Manta Net #1

Source

https://www.sciencedirect.com/science/article/pii/S0165993618305247

https://www.aquaticlivefood.com.au/product/manta-micro-plastics-net/

State the source of the information about this prior design attempt, which should be taken from any of the following credible source categories: peer-reviewed journals (check out your public library for free digital access to journals; here are some free online journals and research databases), professional organizations, government agencies, local and national newspapers, patents (here is a patent search site for the United States Patent and Trademark Office), and reputable vendors and manufacturers. Make sure to incorporate in-text citations in the details section below (i.e. don't just give a list at the end of the element or footnote).

Image

Name: ______________________________ **Date:** _________________ **Class:** _______________

Details

Include main details about this prior design attempt

This product collects surface-level microplastics using a very dense plastic mesh. You attach it to a towline, and the towline to a boat, and drag it along the water surface (It is designed strictly for the water surface). The end of it has a sample collection bottle.

Provide evidence and explain what parts of this prior design attempt were unsuccessful at solving the defined problem and why, based on design requirements. Or describe how you plan to improve on this design.

Analysis

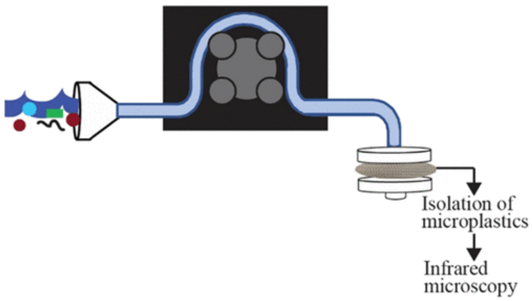
This product is successful in that it can collect very small microplastics, as small as 150um (for the specific product from aquaticlivefood.com). It is also important to note that in general for plankton nets, you can detect as low as 100um, according to ScienceDirect. It also has a wide opening with an exact area, so you can quantify many microplastics in a certain area. The collection bottle at the end also means that we could get many microplastics, but we only have to take a little bit of actual water in our sample. The issue with this design is that it is limited to the surface of the water; it cannot go any deeper, meaning that a huge sampling area is left out. Furthermore, the fact that this net is made of plastic means that it is also leaving behind microplastics as it samples. Not only is this harmful, but it also slightly interferes with the data. Another disadvantage is that this method requires a boat. I would improve on this product by making it usable at any depth, and also a smaller form factor (not requiring a boat).

Prior Design Attempt #2 - Pumps

Source: https://www.sciencedirect.com/science/article/pii/S0165993618305247

Image

Name: ______________________________ **Date:** _________________ **Class:** _______________



Details

Collects samples through fine-mesh filters. Samples large volumes of water. Effortless. Allows choice of mesh size. feature a pump, tubing (non-contaminating), filter holders, and a flow meter to measure the volume of water filtered.

Analysis

Requires equipment; Requires energy to work; Potential contamination by the apparatus; May be difficult to carry between sampling locations. (https://pubs.acs.org/doi/10.1021/acsestwater.1c00270)

Prior Design Attempt #3 - Sieving

State the source of the information about this prior design attempt, which should be taken from any of the following credible source categories: peer-reviewed journals (check out your public library for free digital access to journals; here are some free online journals and research databases), professional organizations, government agencies, local and national newspapers, patents (here is a patent search site for the United States Patent and Trademark Office), and reputable vendors and manufacturers. Make sure to incorporate in-text citations in the details section below (i.e. don't just give a list at the end of the element or footnote).

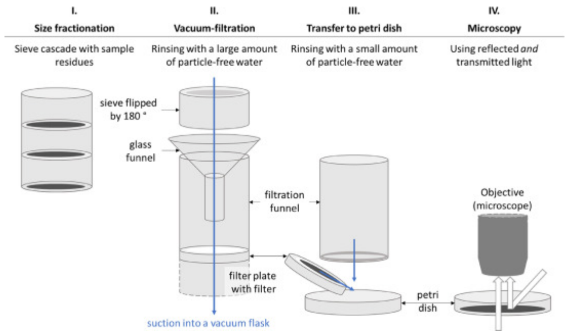
Source

https://www.sciencedirect.com/science/article/pii/S0165993618305247

https://www.sciencedirect.com/science/article/pii/S2215016121001345

Name: ______________________________ **Date:** _________________ **Class:** _______________

Image



Details

Include main details about this prior design attempt

Sieving is a process used when sampling microplastics in water, sand, or sediment. The process of collecting the samples uses a mesh filter to separate rocks and pebbles from finer materials. This process continues when the finer materials pass through more filters, automatically sorting the different sizes of microplastics.

Analysis

The downside to this method is that you must also then sort through all the non-microplastics left behind. This method is a little impractical for an ROV. I would not use this method for our ROV. Perhaps we could, however, use a variety of net/mesh densities to help sort certain-sized microplastics. It is also pretty laborious for humans, so I don't think it would be possible for an ROV to use this method efficiently (sciencedirect.com).